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PI Name:	Oates, Chris Ph.D.		
Project Title:	Development/Evaluation of a 10^-17 Yb Optical Clock for Space Applications		
Division Name:	Physical Sciences		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
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COI Name (Institution):	Diddams, Scott (National Institute of Standards & Technology) Ludlow, Andrew (National Institute of Standards & Technology)		
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Task Description:	The primary goal of the European Space Agency's (ESA) Space Optical Clock (SOC) program is to develop a high-performance optical frequency standard (clock) for operation in space with 1 x 10^-17 frequency uncertainty. National Institute of Standards and Technology (NIST) proposes to leverage its existing research on world-leading Yb lattice optical clocks to: (1) demonstrate a laboratory-based Yb lattice optical clock with frequency uncertainty of 1 x 10^-17 or better; (2) stabilize a space-friendly fiber frequency comb to the Yb clock signal; and (3) develop a preliminary design for space-qualification by providing clock subsystem specifications. The proposed program will provide ESA with detailed information to make the most informed decisions on use of one or more optical frequency standards for the flight portion of the SOC program. Currently, ESA has identified the Sr and Yb lattice optical clocks as the most promising technologies. A Yb lattice clock has significant potential advantages over Sr in that Yb clocks can operate with solid state lasers much more likely to be practical in space operations. Several European groups are presently evaluating Sr's space prospects and Sr has thus received significant attention. For Yb, however, the most mature and high performance lattice clock resides in our lab at NIST. This Yb clock currently performs at least as well as the world's best Sr clocks, approximately 10^-16 frequency uncertainty in all cases, and continues to rapidly improve. Thus, we are uniquely qualified to play a vital role in the SOC project. For a modest investment, the proposed research program will ensure ESA has the best information and options to select the optical clock technology to meet SOC program goals, ensuring fully informed decisions affecting a much larger financial and scientific investment in the SOC program. Furthermore, by demonstrating a 10^-17 level Yb optical lattice clock, our proposed effort will achieve significant milestones currently outlined in the SOC program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NIST SOC program (Yb lattice clock) also directly supports the major benefits of the overall Atomic Clock Ensemble in Space (ACES) program by developing a new high performance NIST optical frequency standard that will directly contribute to NIST's roles in the broader ACES program: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. The NIST Yb lattice clock represents a dramatic improvement in stability and accuracy compared to cesium microwave primary frequency standards. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, search for possible Lorentz transformation variations, and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosomology.
	Progress in the most recent reporting period (April 2016 through March 2017) has been substantial. The two key aspects of improving an atomic clock's performance are to make the ticking more regular (stability) and to make the ticking rate match more closely the reference atom's natural ticking rate (termed uncertainty or accuracy). The major goal of this proposal was to demonstrate a fractional uncertainty of less than one part in 10^17 for the Yb lattice clock. Achieving such an ambitious goal required that we evaluate all the possible effects that could shift the clock frequency away from that of the atom. We note that our continued efforts to improve the stability as well has greatly accelerated this investigation. During the past year, we have completed the exhaustive Yb lattice clock evaluation with a result that is a factor of five improvement beyond the stated goal. The major advances for this period include: 1. Reducing the Key Remaining Clock Shifts The Lattice Light Shift The largest shift for lattice clocks is usually due to the lattice light itself. This shift is a by-product of the trapping light used to confine the Yb atoms in the optical lattice, which is formed by a tightly focused standing wave of light. However, this light also shifts atomic energy levels by as much as 100s of kHz, a huge value given that we are striving to produce clocks with mHz uncertainty. The standard trick that all lattice clock systems use to minimize this effect is to tune the frequency of the lattice light to a value (the so-called "magic wavelength") that equally shifts the clock transition's lower and upper energy levels, thereby leaving the transition frequency unaffected, independent of lattice light intensity to first order. However, to achieve a 10^-18 level for the clock uncertainty, it is necessary to consider higher order light shift effects including the hyperpolarizability (effectively resulting from two-photon transitions starting from the either the lower or upper level) and M1/E2 shifts due to magnetic dipole/electric quadrupole transitions initiating from the clock levels. After performing (and improving) shift measurements with our second Yb clock, which confirmed results from a set of measurements we took with our first clock last year, we then investigated how additional cooling of the atoms would affect the Stark shifts. This research was particularly significant because the thermal influences on lattice Stark effects have never been directly observed in an optical lattice clock. We found some subtle influences that enabled to characterize the clock shifts more precisely. While the effect is relatively small at normal operating conditions, we must account for it as we strive for 10-18 level clock accuracy. These results have been described in a manuscript that has been prepared for journal submission. Finally, in addition to clearing up some of the confusion with the linear- and second-order Stark shifts and reducing

Task Progress:	their contribution to the 10^-18 level, this investigation also gave us a way finally to probe one of the most elusive components of the shift, that due to M1/E2 transitions. In fact, in the literature their contribution to the uncertainty budget of an optical lattice clock at or below the 10^-17 level is debatable, and these effects have never been observed nor measured experimentally for any lattice clock system. Our best estimates of M1/E2 effects are from ab initio atomic structure calculations that we made recently, but theory only gives us an estimate of their magnitude. We took advantage of our sideband cooling capabilities to isolate this effect, since the shift from M1/E2 effects are known to scale strongly with atomic temperature. We expect this first experimental evaluation of this effect to be completed during the current proposal year. Collisions Between Cold Atoms and the Background Gas While our cold, lattice-trapped atoms sit in ultra-high-vacuum, lingering background gas particles at room temperature collide with the trapped atoms. When these comparatively 'hot' background atoms collide head on with our lattice-trapped ytterbium, they eject the Yb atoms from the trap. When they collide instead at a relatively large glancing distance (impact parameter), the ytterbium remains trapped. However, the atom experiences a collision phase shift which differs for electrons in the ground or excited clock states. This causes a frequency shift of the clock transition, which is proportional to the density of background gas atoms, as well as the strength of the interaction between colliding atoms. Estimates predict that this effect could cause a fractional shift larger than 10^-18, making it relevant for high performance lattice clocks. However, no experimental investigation of this effect had been previously carried out in a lattice clock system. So, we proceeded to carry out measurements of two distinct collision processes: (1) hot ytterbium colliding with cold, lattice trapped ytterbium and (2) hot molecular hydrogen colliding with cold, lattice trapped ytterbium. We measured both collisional frequency shifts and trap losses under background gas densities exaggerated from our normal operating conditions, which improved the precision of our shift characterization. We found that the observed background shift, measured over a range of vacuum levels, exhibited clean linear dependence on the collision-induced trap loss rate. This result corroborates existing theory on this dependence and simplifies the characterization of this shift. After a careful study, our measure of this effect was sufficiently precise that we could determine this shift with an uncertainty at the 5 x 10^-19 level. Thus, we now feel, at this point, that this important systematic effect is fully and properly accounted for in our lattice clock. Reducing Remaining Shifts: a Complete Yb Lattice Clock Uncertainty Budget With the studies of the lattice light and background collision shifts completed, we needed only to tie up some loose ends before producing a comprehensive Yb lattice clock uncertainty budget. Most of all, we wanted to be sure that there were no effects due to possible residual motion of the trapped atoms, which could result from atomic tunneling between potential wells of the optical lattice. By performing spectroscopy on the lattice-trapped atoms from alternating directions, we measured residual Doppler effects associated with uncompensated vibrations or phase chirps between the optical lattice and the probe laser. Due to careful stabilization of the phase of the probe laser relative to the reference frame of the trapped atoms, residual Doppler effects are negligible at are current clock performance goals. With the list of systematic effects now complete, we have a preliminary total budget with a fractional frequency uncertainty of ~ 2 x 10^-18. It is now essential that we verify this level of performance through direct frequency comparisons between our two Yb clocks, Yb-1 and Yb-2, to see if their operational clock frequencies indeed agree at the evaluated uncertainty level. To this end, we have constructed a blackbody enclosure, essential for guaranteeing a stable temperature environmental for the trapped atoms, for Yb-2. This enclosure improves upon earlier designs and will allow further reduction of the clock shift uncertainty due to the thermal environment, in the 10^-19 uncertainty range. 2. Further Improvements in Lattice Clock Stability Throughout the duration of this project, we have continually tried to improve the stability of the Yb clock system. Improved stability reduces averaging time and enables us to evaluate clock systematics more rapidly, a critical factor in reaching sub 10^-17 uncertainty for the Yb clock system. During the past year, we have constructed optical reference cavities that should have still lower thermal noise limits (~5 x 10^-17 at room temperature) through the implementation of a new technology based on crystalline mirrors. This reduced noise limit should benefit the pre-stabilization of the clock laser, which in turn should improve the stability of the Yb clock system. Moreover, we finished up research on how one can use two Yb samples to reduce clock instability below one part in 10^16 for one second averaging time, which resulted in a manuscript, "Ultra-stable optical clock with two-cold atom ensembles," which was published in Nature Photonics. A number of news organizations around the nation and abroad highlighted the record atomic clock stability achieved by the ytterbium optical lattice clock. These results also have relevance to making robust and compact optical clocks with state-of-the-art performance.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
Articles in Peer-reviewed Journals	Fortier TM, Rolland A, Quinlan F, Baynes FN, Metcalf AJ, Hati A, Ludlow AD, Hinkley N, Shimizu M, Ishibashi T, Campbell JC, Diddams SA. "Optically referenced broadband electronic synthesizer with 15 digits of resolution." Laser & Photonics Reviews. 2016 Sep:10(5):780-90. http://dx.doi.org/10.1002/lpor.201500307 , Sep-2016
Articles in Peer-reviewed Journals	Schioppo M, Brown RC, McGrew WF, Hinkley N, Fasano RJ, Beloy K, Yoon TH, Milani G, Nicolodi D, Sherman JA, Phillips N, Oates CW, Ludlow AD. "Ultra-stable optical clock with two cold-atom ensembles." Nature Photonics. 2017 Jan:11(1):48-52. http://dx.doi.org/10.1038/nphoton.2016.231 , Jan-2017